Sets, Functions, and the Continuum Hypothesis

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Domain Expansion HIMATIKA ITB



Table of Contents

1. The Notion of Size

2. Countable vs. Uncountable

3. The Role of Axioms

4. The Continuum Hypothesis

5. References

Defining Cardinality

Definitions

- **Finite:** The counting process terminates.
- **Infinite:** Not finite.
- **Countable:** There exists a bijection with the natural numbers $\mathbb{N}$. Its cardinality is $\aleph_0$.
- **Uncountable:** An infinite set that is not countable.

Comparing Sizes with Functions

Two sets M and N have the same **cardinality**, written $|M| = |N|$, if and only if there exists a **bijection** from M to N.

The Cantor-Bernstein Theorem

Theorem (Cantor-Bernstein)
If there exists an injective function from set M into set N, and another injective function from N into M, then there must exist a bijection between M and N.

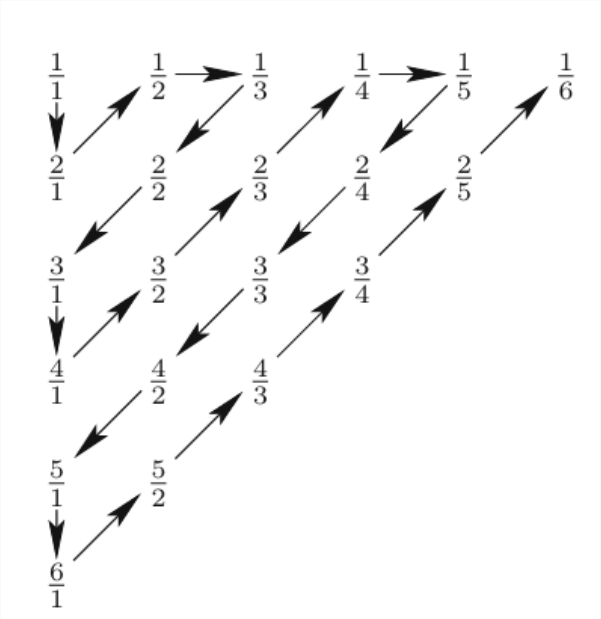
$$(|M| \leq |N| \text{ and } |N| \leq |M|) \implies |M| = |N|$$

This theorem allows us to prove that two sets have the same size without having to explicitly construct a bijection. We just need to find two ”one-way” injections.

The Countability of Rational Numbers ($\mathbb{Q}$)

Method 1: Diagonalization

We can prove $|\mathbb{Q}| = \aleph_0$ by creating an infinite grid of all fractions p/q. By traversing the diagonals, we can create a single list that is guaranteed to hit every rational number.

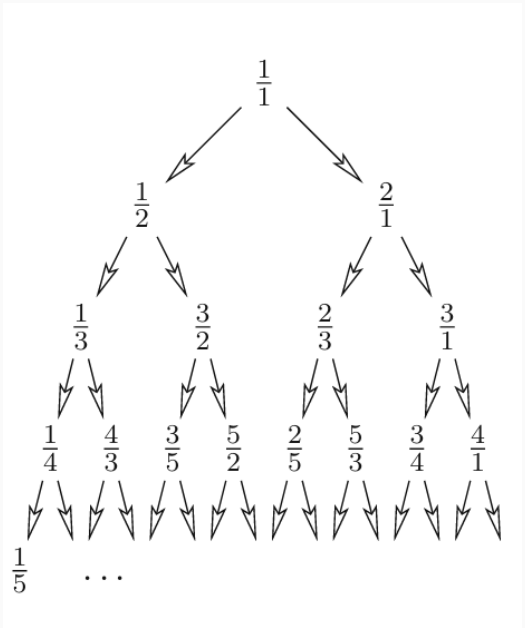


The Countability of Rational Numbers ($\mathbb{Q}$)

Method 2: Calkin-Wilf Tree

More elegant method to list rational numbers using binary tree.

- Start with $\frac{1}{1}$ at the top.
- For any node $\frac{i}{j}$, its left child is $\frac{i}{i+j}$ and its right child is $\frac{i+j}{j}$.



This tree contains every single positive rational number exactly once, in reduced form. $\mathbb{Q}^+$.

The Uncountability of Real Numbers ($\mathbb{R}$)

Theorem
The set of real numbers, $\mathbb{R}$, is uncountable.

Proof by Cantor's Diagonalization
Assume, for contradiction, that we can list all real numbers between 0 and 1.

$$
\begin{aligned}
r_1 &= 0.\mathbf{a_{11}}a_{12}a_{13}\ldots \\
r_2 &= 0.a_{21}\mathbf{a_{22}}a_{23}\ldots \\
r_3 &= 0.a_{31}a_{32}\mathbf{a_{33}}\ldots \\
&\vdots \\
r_n &= 0.a_{n1}a_{n2}a_{n3}\ldots\mathbf{a_{nn}}\ldots \\
&\vdots
\end{aligned}
$$

We can construct a new number $b = 0.b_1b_2b_3\ldots$ such that for every n, the digit b_n is chosen to be different from a_{nn}.

This number b cannot be on the list, as it differs from every number r_n in at least the n-th decimal place. This is a contradiction, proving the list is incomplete.

We denote the cardinality of the continuum (the real numbers) as c. We have shown $\aleph_0 < c$.

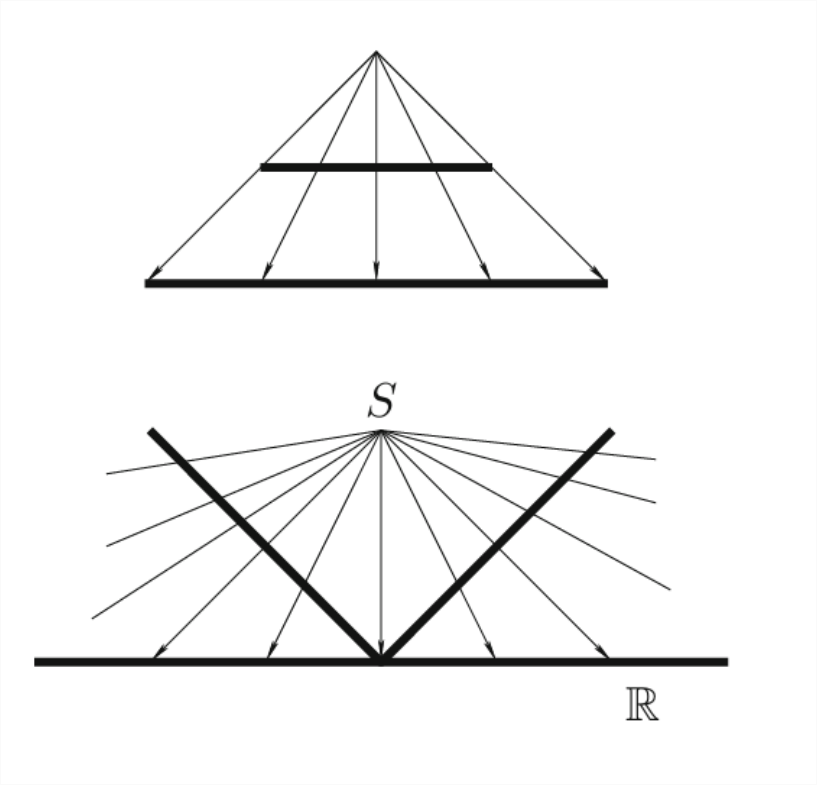


Figure 1: Projection from subinterval of $\mathbb{R}$ to $\mathbb{R}$

Cardinality of the Plane ($\mathbb{R}^2$)

Theorem
The 2D plane has the same cardinality as the 1D line: $|\mathbb{R}^2| = |\mathbb{R}|$.

Proof Sketch
We create a bijection from the unit square $(0, 1] \times (0, 1]$ to the unit interval $(0, 1]$. First, group the digits of x and y by their non-zero entries:

$$x = 0.3\ 01\ 2\ 007\ 08\ \ldots$$
$$y = 0.009\ 2\ 05\ 1\ 0008\ \ldots$$

Then, construct a new number z by alternating these groups to form a single decimal expansion:

$$z = 0.300901220500710800 08\ldots$$

This mapping is a bijection, which proves that $|\mathbb{R}^2| = |\mathbb{R}|$.

The Axiom of Choice (AC)

The Axiom of Choice
For any collection of non-empty sets, it is possible to construct a new set by choosing one element from each set in the collection.

The Well-Ordering Theorem
The Axiom of Choice is logically equivalent to the Well-Ordering Theorem, which states: **Every set can be well-ordered.**

What is a Well-Ordering?
An ordering where every non-empty subset has a least element. $\mathbb{N}$ is well-ordered; $\mathbb{Z}$ and $\mathbb{R}$ are not under their usual orderings. AC allows us to impose such an order on them.

The Next Cardinal

Because every set can be well-ordered (implied from AC), we can arrange all implied from cardinalities into a sequence.

Theorem
*For every cardinal number $\mathfrak{m}$, there is a definite **next larger** cardinal number.*

This allows us to create the sequence of aleph numbers:

$$\aleph_0 < \aleph_1 < \aleph_2 < \cdots < \aleph_\omega < \dots$$

$\aleph_1$ is, by definition, the very first cardinality larger than $\aleph_0$.

The Continuum Hypothesis (CH)

We have established two key facts:

1. The cardinality of the real numbers is c. We know $\aleph_0 < c$.
2. The first infinity after $\aleph_0$ is $\aleph_1$.

The Continuum Hypothesis
The hypothesis is the simple assertion that these two values are the same:

$$c = \aleph_1$$

It claims there is no set with a size strictly between that of the integers and that of the real numbers.

An Independent Truth

The Status of CH

- **Kurt Gödel (1940):** Proved that CH **cannot be disproved** from the standard axioms of set theory (ZFC). One can consistently assume it is true.
- **Paul Cohen (1963):** Proved that CH **cannot be proved** from ZFC. One can also consistently assume it is false.

Conclusion

The Continuum Hypothesis is **independent** of the ZFC axioms. Its truth value is a choice, leading to different ”universes” of mathematics.

References

 Aigner, Martin and Günter M. Ziegler (2018). ***Proofs from THE BOOK.*** 6th. Berlin, Heidelberg: Springer-Verlag.

 Cohen, Paul J. (1966). ***Set Theory and the Continuum Hypothesis.*** New York: W. A. Benjamin.

Thank you.